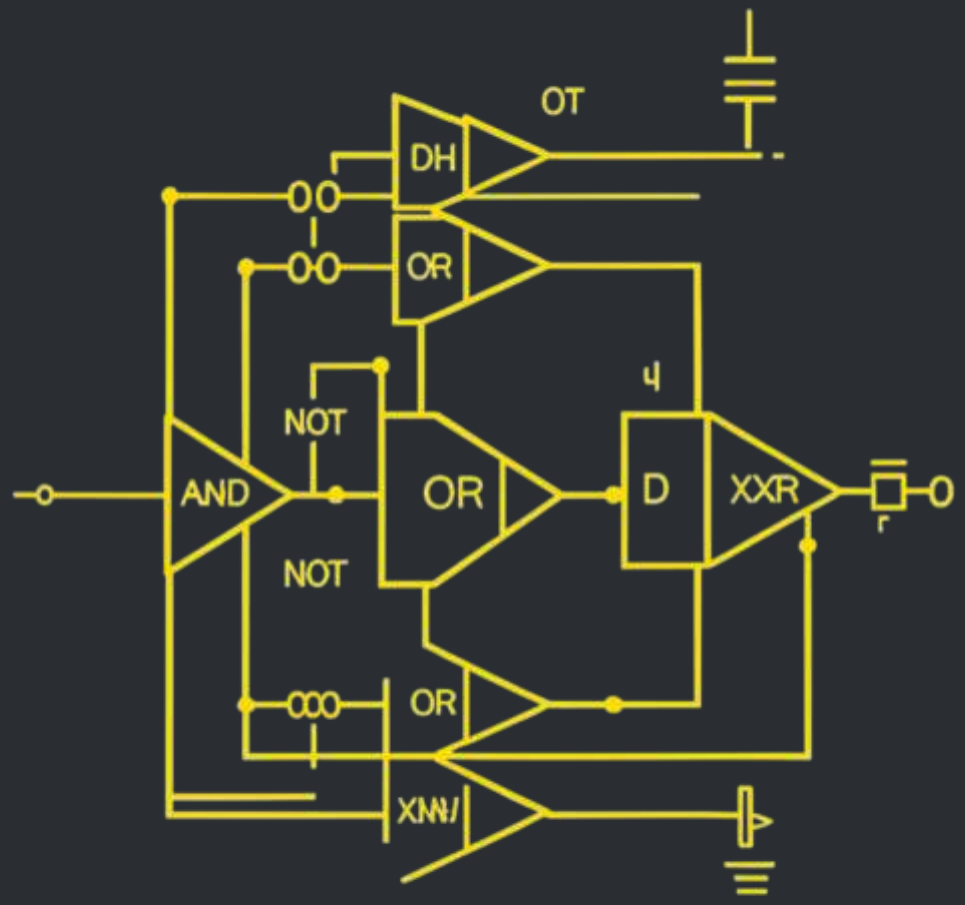
DIGITAL LOGIC · CLASSROOM PROJECT

Design and Simulation of a Digital Alert System Using Logisim

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INTRODUCTION

Introduction and Objective

A digital alert system monitors multiple input conditions and activates an alert when at least one specified condition is triggered.

Each monitored input is a binary signal (0 or 1). Because the alert depends only on the current input values and stores no state, it is a combinational logic circuit: its objective is to map every input combination to the correct output.

01

Design

Design a simple digital alert circuit.

02

Apply

Apply basic logic gates to process inputs.

03

Verify

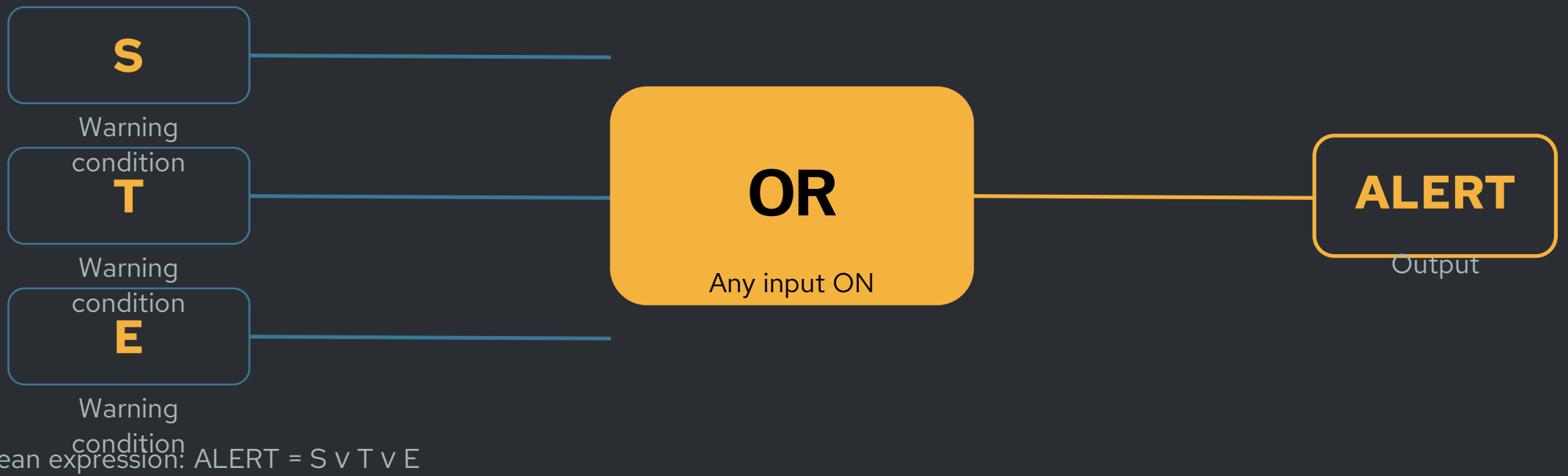
Simulate and verify the circuit using Logisim.

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THEORY · LOGIC CIRCUIT

Theory and Logic Circuit

Inputs represent different warning conditions. Binary values 0 and 1 represent OFF and ON. An OR gate activates the output if any input is ON.



Boolean expression: ALERT = S v T v E

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IMPLEMENTATION

Circuit Design in Logisim

When any input is switched ON, the alert output becomes 1. When all inputs are OFF, the output remains 0.

01

Inputs

Three input switches: S, T and E.

02

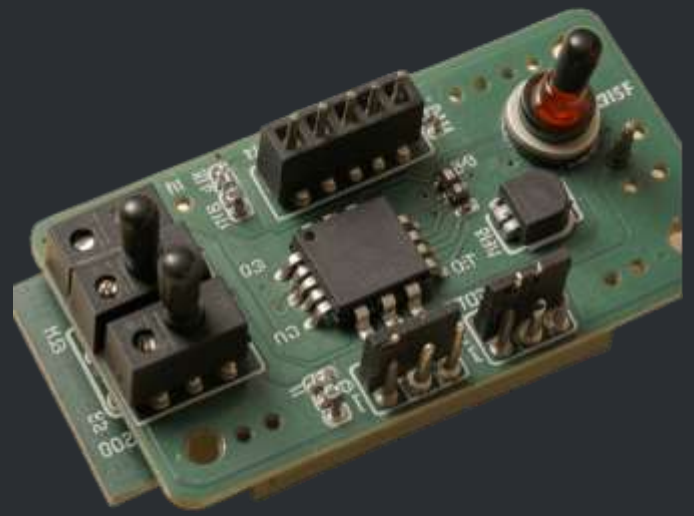
Logic

One OR gate to combine the inputs.

03

Output

One output LED or pin labeled ALERT.



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SIMULATION

Simulation Results

S	T	E	ALERT
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1



All inputs OFF: alert OFF. One input ON: alert ON. All eight input combinations are verified; only S=0, T=0, E=0 has alert OFF.

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CLOSING

Conclusion and Applications

The project demonstrates how basic digital logic can be used to design and simulate an alert system.

Combinational alert logic is useful because the output depends only on the current inputs, so an OR-gate network responds immediately when any monitored condition is active. Its limitations are that it has no memory of past events and no adjustable thresholds. Additional inputs or gates can extend the design by widening the OR expression to more sensors or conditions.



Security monitoring



Temperature-warning systems



Emergency indication systems



Future scope

add more inputs or additional logic conditions.

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